

Rajalakshmi Engineering College

Name: Harivarsha s
Email: 240801109@rajalakshmi.edu.in
Roll no:
Phone: 6380939059
Branch: REC
Department: I ECE FA
Batch: 2028
Degree: B.E - ECE

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 7_COD_Question 1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Ravi is building a basic hash table to manage student roll numbers for quick lookup. He decides to use Linear Probing to handle collisions.

Implement a hash table using linear probing where:

The hash function is: $\text{index} = \text{roll_number} \% \text{table_size}$ On collision, check subsequent indexes (i+1, i+2, ...) until an empty slot is found.

You need to:

Insert a list of n student roll numbers into the hash table. Print the final state of the hash table. If a slot is empty, print -1.

Input Format

The first line of the input contains two integers n and table_size, where n is the

number of roll numbers to be inserted, and table_size is the size of the hash table.

The second line contains n space-separated integers — the roll numbers to insert into the hash table.

Output Format

The output should print a single line with table_size space-separated integers representing the final state of the hash table after all insertions.

If any slot remains unoccupied, it should be represented as -1.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 4 7

50 700 76 85

Output: 700 50 85 -1 -1 -1 76

Answer

```
#include <stdio.h>
```

```
#define MAX 100
```

```
#include <stdio.h>
```

```
#define MAX 100
```

```
void initializeTable(int table[], int table_size) {  
    for (int i = 0; i < table_size; i++)  
        table[i] = -1;  
}
```

```
void insertIntoHashTable(int table[], int table_size, int arr[], int n) {  
    for (int i = 0; i < n; i++) {  
        int index = arr[i] % table_size;  
  
        while (table[index] != -1) {  
            index = (index + 1) % table_size;
```

```

    }
    table[index] = arr[i];
}
}

void printTable(int table[], int table_size) {
    for (int i = 0; i < table_size; i++)
        printf("%d ", table[i]);
    printf("\n");
}

int main() {
    int n, table_size;
    scanf("%d %d", &n, &table_size);

    int arr[MAX];
    int table[MAX];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    initializeTable(table, table_size);
    insertIntoHashTable(table, table_size, arr, n);
    printTable(table, table_size);

    return 0;
}

```

Status : Correct

Marks : 10/10